Module 2: Impacts of Animal Agriculture: Part 1

Climate, health, and environment
Week 1 | Online-Synchronous | Friday

This session examines how animal agriculture shapes some of the most pressing global and national challenges of our time: climate change, public health, and environmental degradation. Fellows are introduced to the ways livestock production contributes to greenhouse gas emissions, land and water use, biodiversity loss, and pollution, as well as its links to diet-related disease and zoonotic risk.

The session emphasizes that these impacts are not incidental, but are produced by structural features of prevailing food systems. Participants are encouraged to think critically about trade-offs, responsibility, and the implications of these impacts for Just Food Systems in the Philippine context.

Resources[1]

For this session, you will engage with materials that explain the environmental and health impacts of animal agriculture at both global and Philippine scales.

A. Climate and Environment

1. **Intergovernmental Panel on Climate Change – AR6 Working Group III: Agriculture, Forestry, and Other Land Use**
   This resource outlines the role of livestock and food systems in global greenhouse gas emissions and land-use change. As you read, focus on how animal agriculture compares with other emission sources and what mitigation pathways are identified.

2. **Food and Agriculture Organization of the United Nations – Livestock's Long Shadow (Executive Summary)**
   This foundational FAO report explains the environmental impacts of livestock production, including climate change, water pollution, land degradation, and biodiversity loss. Consider why these impacts persist despite long-standing awareness.

B. Health

1. **Bryant, C. J. (2023). Plant-based animal product alternatives are healthier and more environmentally sustainable than animal products.**
   This review synthesizes 43 studies comparing animal products and plant-based animal product alternatives (PB-APAs). As you read, focus on evidence related to nutritional profiles, weight management, disease risk, and how PB-APAs may displace animal products in real-world diets.

2. **Hemler, E. C., & Hu, F. B. (2023). Plant-Based Diets for Personal, Population, and Planetary Health.**
   This paper situates plant-based diets within a unified framework of personal health, population-level disease prevention, and environmental sustainability. Pay attention to how shifts away from animal-heavy diets are associated with lower risks of obesity, type 2 diabetes, cardiovascular disease, and certain cancers, and how these benefits scale beyond individual choice.

Reflection[2]

As you engage with these resources, consider:

- How animal-based diets affect health directly, not just abstractly;
- How plant-based diets and alternatives differ in nutritional outcomes, not only ethics; and

Why health-based arguments can be especially powerful—and sometimes contentious—in food systems advocacy.